

illuminati - 2019

Advanced Physics Assignment-4B | JEE-2019

Modern Physics & Optics

- A beam of ultraviolet light of all wavelength passes through hydrogen gas at room temperature, in the x -direction. Assume that all photons emitted due to electron transitions inside the gas emerge in the y -direction. Let A and B denote the lights emerging from the gas in the x -and y -directions respectively :

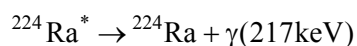
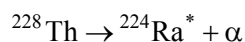
 - Some of the incident wavelengths will be absent in A
 - Only those wavelengths will be present in B which are absent in A
 - B will contain some visible light
 - B will contain some infrared light
- A neutron collides head-on with a stationary hydrogen atom in ground state. Which of the following statements are correct : (Assume that the hydrogen atom and neutron has same mass)

 - If kinetic energy of the neutron is less than 20.4 eV collision must be elastic
 - If kinetic energy of the neutron is less than 20.4 eV collision may be inelastic
 - Inelastic collision may take place only when initial kinetic energy of neutron is greater than 20.4 eV
 - Perfectly inelastic collision can not take place
- Energy distribution in a nuclear reaction: In the nuclear reaction ${}_0^1n + {}_5^{10}B \rightarrow {}_3^7Li + {}_2^4He$ a slow moving neutron ($K_n \rightarrow 0$) strikes a boron at rest. After reaction He is observed to have a speed of $9.3 \times 10^6 \text{ m/s}$. Find the kinetic energy of Lithium and Q value of reaction. Given masses are $M_{He} = 4.0026 \text{ amu}$; $M_{Li} = 7.0160 \text{ amu}$.
- Interaction force between fission fragments: In the fission of ${}_{94}^{239}Pu$ by a thermal neutron, two fission fragments of equal masses and sizes are produced and 4 neutrons are emitted. Find the force between the two fission fragments at the moment they are produced. Take fermi constant $R_0 = 1.1 \text{ fermi}$.
- Gamma decay followed by beta: ${}^{57}Co$ decays to ${}^{57}Fe$ by β^+ emission. Thus resulting ${}^{57}Fe$ is in its excited state and comes to ground state by emitting γ - rays. The half life of β^+ decay is 270 days and that of γ emission is 10^{-8} seconds. A sample of ${}^{57}Co$ gives 5×10^9 gamma photons/sec. Find the time after which rate of γ - photon emission drops to 1.25×10^9 photons/sec.
- Photoelectric effect from a metal sphere: A point source of light of power 3.2 mW emits monochromatic light photons of energy 5 eV. The source is located at a distance of 0.8 m from the centre of a stationary metallic sphere of work function 3eV and radius $8 \times 10^{-3} \text{ m}$. The efficiency of photo electric emission is one for every 10^6 photons incident on sphere. Assume all ejected electrons are instantly swept away. Find the time after which emission stops.

7. De-Broglie wavelength in frame of centre of mass : Two identical non relativistic particles move at right angle to each other, possessing De-Broglie wavelengths λ_1 and λ_2 . Find the De-Broglie wavelengths of both particles in reference frame attached to their centre of mass.
8. Photoelectrons in chamber of α – particles: When a $4000\text{\AA}$ radiation is incident on a metal surface of work function 1.9eV . Photoelectrons with maximum kinetic energy are emitted which are allowed to pass through a chamber containing α – particles. These electrons combine with α – particle to reduce He^+ ions in fourth excited state. Find the energies of the photons, lying in 2 to 4 eV range, that are likely to be emitted during and after the combination.
9. Revolution of a π – meson : A pi-meson hydrogen atom is a bound state of negatively charged pion and a proton. A pion has a mass 273 times that of electron with $-e$ charge. If average life of a π – meson is 10^{-8} second, find the number of revolution made by π – meson in the ground state of the atom before it decays.
10. Atoms colliding a Heavy Target: Two hydrogen like atoms A and B having different masses and equal number of protons and neutron in each when moving with same velocity and strikes a heavy target rebound back with same velocities. In this process B imparts twice the momentum to the target than that A imparts. If the energy difference between first line of Balmer series emitted by A and B is 5.667 eV identify the atoms A and B.
11. ✓ Simultaneous Decay of Alpha and Beta Emission: A radio isotope ^{212}Bi decays by two modes of α and β emission and respective probability ratio is $\frac{7}{13}$. The half life of ^{212}Bi is one hour. If 1 gm of pure ^{212}Bi is taken at 12:00 Noon, find the composition of sample at 1 PM.
12. Age Determination by Radioactivity Series: In nature a decay chain series starts with $_{90}\text{Th}^{232}$ and finally terminates at $_{82}\text{Pb}^{208}$. A thorium sample was found to contain $8 \times 10^{-5}\text{ ml}$ of helium at STP and $5 \times 10^{-7}\text{ gm}$ of $_{90}\text{Th}^{232}$. Find the age of sample assuming the source of helium to be only the decay of Th^{232} . Also assume complete retention of helium within the ore sample. Take half life of Th^{232} to be 1.39×10^{16} years.
13. Stationary Wave Formation by De Broglie Waves of Electron : In a lattice atoms are arranged in an one dimensional array. Assume that the de-Broglie wave associated with an electron can form a standing wave between the atoms with nodes at each atom. It is found that such a stationary wave is obtained when the distance between atoms is $2\text{\AA}$ and a similar such stationary wave is again formed if the separation increases to $2.5\text{\AA}$ and not for any intermediate separation. Find the energy of electron and minimum separation between atoms for which such a wave can be obtained.
14. Potassium Collecting Energy : A potassium surface is placed 75 cm away from a 100 W bulb. It is found that the energy radiated by the bulb is 5% of the input power. Consider each potassium atom as a circular disc of diameter $1\text{\AA}$. Determine the time required for each atom to absorb an amount of energy equal to its work function 2 eV. Also find the time if we consider atom to be spherical.
15. Photoelectrons Going Around an Ion : A small metal plate of work function ϕ and it is kept at a distance d from a singly ionized, fixed ion. A monochromatic light beam is incident on the metal plate and photoelectrons are emitted. Find the maximum wavelength of the light beam so that some of the photoelectrons may go round the ion along a circle.

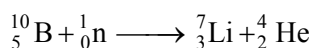
16. Reaction Induced by α – particle: An α – particle with kinetic energy k initiates a nuclear reaction ${}^9\text{Be}(\alpha, n){}^{12}\text{C}$ with energy yield Q . Find the kinetic energy of the neutron outgoing at right angles to the motion direction of the α – particle. Take masses of neutron, α – particle and carbon as m_n, m_α and m_c .

17. Gamma Emission of Alpha Decay: ${}^{232}\text{Th}$ emits an alpha particle and reduce to ${}^{224}\text{Ra}$. ${}^{224}\text{Ra}$ thus obtained in excited state which comes to ground state by emission of a gamma photon according to reaction.



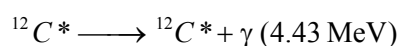
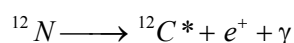
Atomic masses are: ${}^{232}\text{Th} = 228.028726$ amu; ${}^{224}\text{Ra} = 224.020196$; ${}^4\text{He} = 4.00260$ amu . Find the kinetic energy of the emitted α – particle.

18. Alpha Emission in an Induced Reaction: Thermal neutrons (Energy = 0.04 eV) are used to induce the reaction :



Emitted α – particles in this reaction have energy 1.83 MeV, Given the masses of boron, neutron and ${}^4_2\text{H}$ are 10.01167 amu, 1.00894 amu and 4.00386 amu respectively. Find the mass of Lithium.

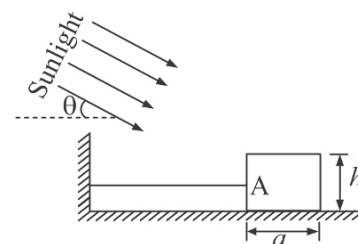
19. Maximum kinetic energy of a β particle : Calculate the maximum kinetic energy of the beta particle emitted in the following decay process:



Take atomic mass of ${}^{12}\text{N} = 12.018613$ amu; ${}^{12}\text{C} = 12.00000$ amu & $m_e = 0.00055$ amu

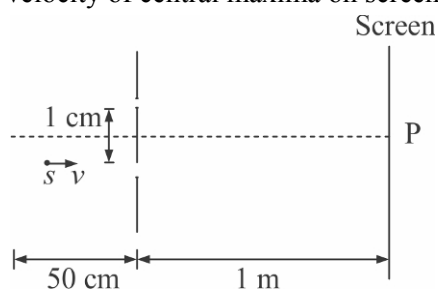
20. Energy of Gamma Emission : It is observed that ${}^{210}\text{Po}$ nuclei (in ground state) emits two group of α -particles with kinetic energies 5.30 MeV and 0.50 MeV – following the emission of these particles the daughter nuclei are found in the ground state and excited states. Find the energy of gamma photon emitted by the excited nuclei.

21. Tension in a string due to radiation of sunlight : Figure shows a block A of mass m (size $a \times b \times h$) placed on a horizontal smooth surface attached to a side wall by a light string. If sun light of intensity I falls on it at an angle θ to horizontal and surfaces of block are perfectly black, find tension in connecting string.

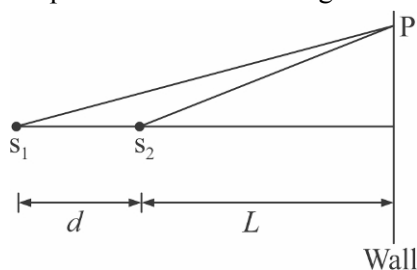


22. Photoemission by Black Body Radiation : A black body emits peak spectral intensity at $9000 \text{ \AA}$. On increasing its temperature total radiation power increases by 81 times. At the initial temperature when peak radiation of black body is allowed to fall on a metal surface (assume other radiation are blocked by a filter), it does not cause any photoemission but after increasing temperature the peak radiation of black body causes electrons come out which are stopped by a potential equal to the excitation energy of H-atoms for $n = 2$ to $n = 3$ Bohr level. Find work function.

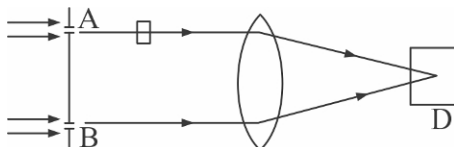
23. De-excitation of a moving hydrogen atom : Consider a hydrogen atom with its electron excited to n^{th} state moving at speed v ($\ll c$). It emits a photon in the direction of its motion and changes its state to a lower state m . If atom is kept at rest then for the transition from n to m state a frequency ν_0 is emitted, find the frequency of emitted photon in case when atom is moving.
24. De-Excitation of an Excited H-Like Gas : Atoms of hydrogen like gas are in a particular excited energy level. When these atoms de-excite they emit photons of different energies. Maximum and Minimum energies of emitted photons are 52.224 eV and 1.224 eV respectively. Identify the gas and principal quantum number of the initially excited level.
25. Movement of fringe pattern in YDSE : In a YDSE setup a point source of monochromatic light is kept as shown in the figure. If the source is given an instantaneous velocity 1 mm per second toward screen, find the velocity of central maxima on screen.



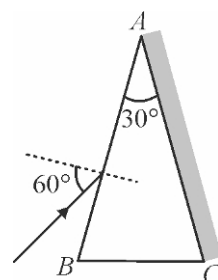
26. No Intensity Points on a Wall : Figure shows two point sources S_1 and S_2 separated by a distance $d = 3\lambda$. Both sources are placed at a distance L from the wall with $d \ll L$ and are having equal intensities. Find all points on wall where light intensity will be zero.



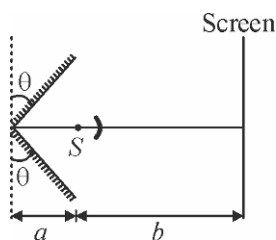
27. Power Collection at Focus of a Convex Lens : A parallel beam of light ($\lambda = 6000 \text{ \AA}$) and intensity $(10/\pi) \text{ W/m}^2$ is incident normally on two circular apertures A and B of radii 0.001 m and 0.002 m respectively. A transparent film of thickness $2000 \text{ \AA}$ and refractive index 1.5 is placed in front of aperture A . Calculate the power reception by a detector D placed at focal spot of the lens. Assume 10% of the power from each aperture is transmitted through lens.



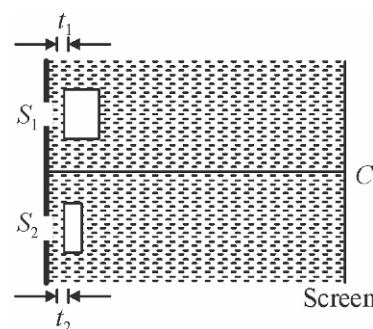
28. A light wavelength $6600 \text{ \AA}$ is incident on face AB of a prism of prism angle 30° at an angle of incidence 60° . On face AC of prism a coating of transparent material ($\mu = 2.2$) is done. Find minimum thickness of the film coating for which emerging light will have minimum intensity. Given that refractive index of prism material is $\sqrt{3}$.



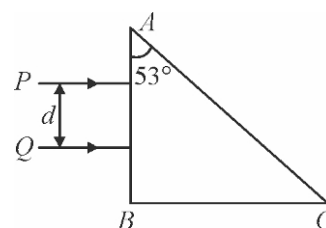
29. Two plane mirrors and a monochromatic source of light are arranged as shown in figure. The angle θ is very small. Find the fringe width of interference pattern obtained on screen placed as shown. Take light wavelength λ .



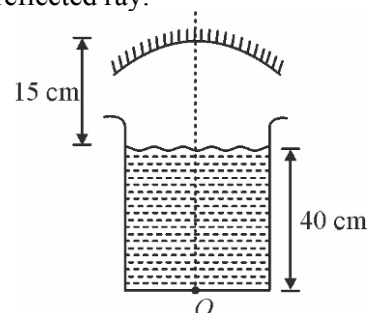
30. In a YDSE setup, screen is placed at a distance 80 cm from slit plane having slits S_1 and S_2 at a distance 2 mm apart. Slit S_1 is covered by a transparent sheet of thickness $t_1 = 2.5\mu\text{m}$ and S_2 is covered by similar sheet of thickness $t_2 = 1.25\mu\text{m}$. Refractive index of both sheet material is 1.4. Now water is filled between slit plane and screen ($\mu_w = 4/3$) and a lighter of wavelength $5000\text{\AA}$ incident normally on slit plane. Find ratio of light intensity at screen centre C to the maximum intensity on screen. Assume intensity of light on screen due to both slits is equal.



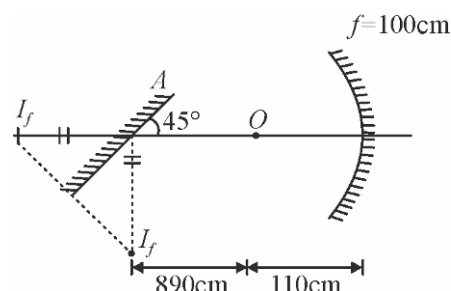
31. Two parallel beams of light P and Q containing radiations of wavelength $4000\text{\AA}$ and $5000\text{\AA}$ which are mutually coherent are incident a prism normally as shown in figure. The refractive index of prism is given as $\mu = 1.20(b/\lambda^2)$ where λ is in $\text{\AA}$ and b is a constant. Calculate value of b so that only one wavelength suffers total internal reflection at face AC . Other wavelength when comes out from AC will have intensities $4I$ and I in beams P and Q . These emerging beams are brought to focus by a convex lens. Find light intensity at focus.



32. A ray of light incident on a plane mirror along a vector $\hat{i} + \hat{j} - \hat{k}$. The normal on point of incidence is along vector $\hat{i} + \hat{j}$. Find the unit vector along the direction of reflected ray.
33. A concave mirror is placed over a beaker filled with water ($\mu = 4/3$) upto 45 cm as shown. The final image of an object placed at bottom of beaker is formed at a distance 16 cm below the water level. Find the focal length of mirror.

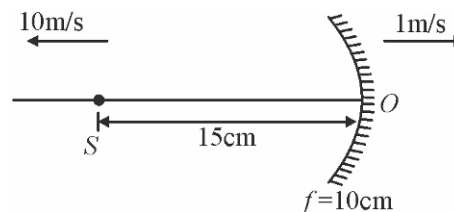


34. Figure shows a concave mirror and an inclined plane mirror at a separation 1000 cm and a point object O placed on topic axis. Find the image of object after two successive reflections taken first at concave mirror.



35. On a large horizontal flat mirror facing upward an equilateral prism is placed with $\mu = \sqrt{2}$. A light passes through the prism symmetrically and after emergence it is incident on the mirror. Find the total deviation of the light ray.

36. A concave mirror of radius of curvature 20 cm and a point source of light in front of it on its principle axis are at separation 15 cm. If both are moving at speed 1 m/s and 10 m/s respectively away from each other as shown, find the velocity of image.



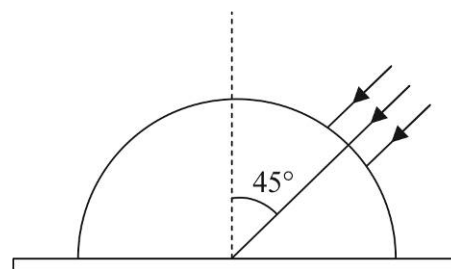
37. Image Produced by a Thick Mirror : An object is placed in front of a glass slab ($\mu = 1.5$) of thickness 6 cm at a distance 28 cm from it. Other face of glass slab is silvered. Find the position of final image of object.

38. Equation of Trajectory of Light Ray: a ray of light incident on a glass slab at grazing incidence. The refractive index of the material of the slab is given by $\mu = \sqrt{1+y}$. If the thickness of the slab is d , determine the equation of trajectory of the ray inside the slab and coordinates of the point from which ray will exist from slab. Take origin at the point of incidence.

39. Maximum range of vision on Ground: Due to a temperature gradient in atmosphere, in a desert, refractive index of air varies with height y as $\mu = \mu_0 \sqrt{1+ay}$ where μ_0 is refractive index at ground and a is a positive constant. Find the maximum distance on ground a man of height h can see in front of him.

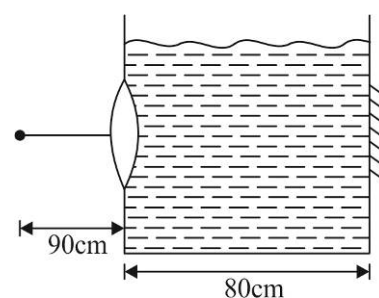
40. Grazing Incidence on a Right-angled Prism: a prism has prism angle 90° . Given that α is the angle of minimum deviation by prism and β is the deviation of ray which enters the prism at grazing incidence. Prove that $\sin \alpha = \sin^2 \beta$.

41. A Light Beam Normally Incident on a Hemisphere: Figure shows a glass hemisphere ($\mu = 2$) is placed on a horizontal surface. A narrow light beam normally incident on the surface of sphere as shown. Find where the final image will be produced by the light beam.

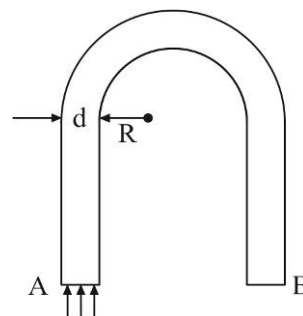


42. Achromatic Doublet of Two Lenses: The dispersive power of crown glass and flint glass are 0.03 and 0.05 respectively. It is desired to make an achromatic doublet with an effective focal length 20 cm. What should be the focal length of the two lenses used for this purpose.

43. A Lens at a Tank Wall: Figure shows a tank of width 80 cm which on one side of the wall has an equiconvex lens ($\mu = 3/2$) focal length 30 cm (in air) fixed. Tank is filled with water ($\mu = 4/3$) and on other side a plane mirror is attached in front of lens. Find where the final image of an object is produced by this system which is placed at a distance 90 cm on the axis of lens.

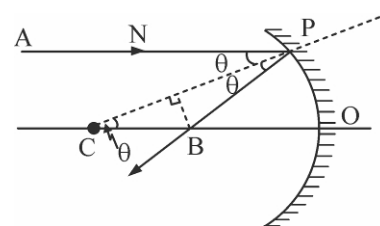


44. Internal Reflection inside a Bent Rod : Figure shows a rod of diameter d bent in shape of an arc of inner radius R as shown. Refractive index of its material is $3/2$. Find maximum value of d/R for which all light rays entering normally at face A of rod will come out only from end B.



45. Light Energy Escaping The surface of water: A point source of light is placed a distance h below the surface of a large deep lake. What fraction of light will escape through the surface of water. Take refractive index of water μ .

46. Figure shows a light ray A, parallel to principal axis incident on a concave mirror at an angle of incidence θ . After reflection at this point light ray passes through axis at point B. Find distance CB. Also find CB is $\theta \rightarrow 0$.



47. A recent high-precision determination of g has a quoted error of 6 parts in 10^9 . Estimate the increase in height at the Earth's surface which gives a change in g equal to this error. If the dependence of g on geographical latitude at sea level is given by

$$g = g_0(1 + \beta \sin^2 \phi),$$

Where ϕ is the latitude and β is dimensionless constant with a value of 0.0053, estimate the northward displacement near latitude 45° which gives a change in g equal to the quoted error.

48. In an experiment to determine the charge-to-mass ratio e/m of the electron using a cylindrical diode, the following equation is derived :

$$\frac{e}{m} = \frac{8V}{r^2 B^2},$$

where V is the potential difference across the diode of radius r at the critical magnetic field B . The latter is supplied by passing a current I through a solenoid of diameter D and length L , and is given by :

$$B = \frac{\mu_0 n I}{(1 + D^2/L^2)^{1/2}}$$

where n is the number of turns per unit length (see problem) The appropriate experimental quantities are determined as follows :

$$n = 3920 \text{ m}^{-1};$$

$$D = 0.035 \pm 0.001 \text{ m};$$

$$L = 0.120 \pm 0.001 \text{ m};$$

$$r = (3.4 \pm 0.1) \times 10^{-3} \text{ m};$$

$$I = 1.92 \pm 0.02 \text{ A};$$

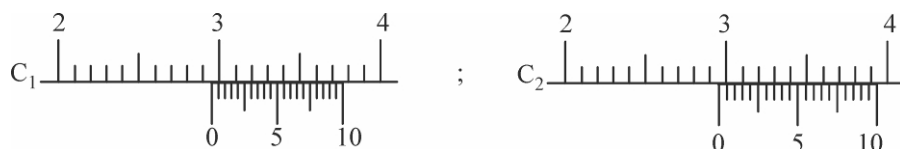
$$V = 20 \pm 1 \text{ V}$$

- (i) Determine the magnitude of the field B and the error in this quantity.
- (ii) Determine the value of e/m and the error in this quantity.

49. The diameter of a cylinder is measured using a Vernier calipers with no zero error. It is found that the zero of the Vernier scale lies between 5.10 cm and 5.15 cm of the main scale. The Vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of the Vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is :

(1) 5.112 cm (2) 5.124 cm (3) 5.136 cm (4) 5.148 cm

50. There are two Vernier calipers both of which have 1 cm divided into 10 equal divisions on the main scale. The Vernier scale of one of the calipers (C_1) has 10 equal divisions that correspond to 9 main scale divisions. The Vernier scale of the other caliper (C_2) has 10 equal divisions that correspond to 11 main scale divisions. The readings of the two calipers are shown in the figure. The measured values (in cm) by calipers C_1 and C_2 , respectively, are :



(1) 2.87 and 2.86 (2) 2.85 and 2.82 (3) 2.87 and 2.87 (4) 2.87 and 2.83

51. A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and that the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides with the main scale line?

(1) 0.50 mm (2) 0.75 mm (3) 0.80 mm (4) 0.70 mm

52. Consider a Vernier calipers in which each 1 cm on the main scale is divided into 8 equal divisions and a screw gauge with 100 divisions on its circular scale. In the Vernier calipers, 5 divisions of the Vernier scale coincide with 4 divisions on the main scale and in the screw gauge, one complete rotation of the circular scale moves it by two divisions on the linear scale. Then:

- (A) If the pitch of the screw gauge is twice the least count of the Vernier calipers, the least count of the screw gauge is 0.01 mm.
 (B) If the pitch of the screw gauge is twice the least count of the Vernier calipers, the least count of the screw gauge is 0.005 mm.
 (C) If the least count of the linear scale of the screw gauge is twice the least count of the Vernier calipers, the least count of screw gauge is 0.01 mm.
 (D) If the least count of the linear scale of the screw gauge is twice the count of the Vernier calipers, the least count of the screw gauge is 0.005 mm.